5B Expanding Brackets and Factorising Linear Expressions

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Expand each bracket using the distributive law:

(a) $5(2x - 7)$ (b) $-4(3a + 5)$ (c) $-6(4 - 3m)$ (d) $3x(4x - 5y)$

Q2. Expand and simplify by collecting like terms:

(a) $4(3x + 2) + 5(2x - 3)$ (b) $6(2a - 1) - 3(3a + 4)$ (c) $5(4m - 3) - 2(6m - 5)$ (d) $2x(3x - 4) - x(5x - 7)$

Q3. Factorise each expression completely by extracting the Highest Common Factor:

(a) $15x + 25$ (b) $18a - 24b$ (c) $14m^2 - 21m$ (d) $16x^2y + 24xy^2$

Q4. Factorise each expression by extracting a negative common factor:

(a) $-8x - 20$ (b) $-18a + 12$ (c) $-15m^2 - 25m$ (d) $-21p^2q + 14pq$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A rectangular swimming pool has width $8\ \text{m}$ and length $(3x + 5)\ \text{m}$. Write the area of the pool in (a) factorised form and (b) expanded form.

Q6. A school orders 6 sets of robotics kits, each containing $2x$ sensors and 5 motors, and returns 2 damaged kits that each contained $(x - 3)$ sensors. Expand and simplify $6(2x + 5) - 2(x - 3)$.

Q7. The area of a rectangular banner is $(28x^2 - 35x)\ \text{cm}^2$. If the width of the banner is the Highest Common Factor of the two terms, find the width and the length of the banner.

Q8. Three sides of a scalene triangle have lengths $3(2x + 1)\ \text{cm}$, $4(x + 3)\ \text{cm}$, and $2(3x - 5)\ \text{cm}$. Expand and simplify to find the perimeter of the triangle in expanded form and then in factorised form.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. When expanding $14 - 3(2x - 5)$, Toby first subtracts $14 - 3 = 11$ and writes $11(2x - 5) = 22x - 55$. Explain why Toby's first step violates the order of operations and find the correct simplified expression.

Q10. Prove algebraically that for any integer n, the expression $5(2n + 3) - 3(2n - 1) + 2(n - 3)$ is always a multiple of 6 by expanding, collecting like terms, and factorising out 6.